MoonFall: Final Mission

Astrobee: Hello Rookie! You are the newly appointed commander of MoonFall Outpost, located on the elevated rim of Shackleton Crater near the lunar south pole. You're responsible for every final decision. The four crew members provide technical knowledge, report problems, and assist with tasks, but they never make the final decision.

Scene-1 : Commandor is inside the command module, checking the outpost's morning systems. AstroBee suddenly interrupts.

Mission 1 — The 30-Minute Warning 05:30

Astrobee: "Good morning, Commander Commandor. I have detected an unusual solar event."
"Our solar observatory has identified a major Solar Particle Event. High-energy particles are approaching the Moon."
"Estimated arrival: 30 minutes. We must immediately activate the emergency protocol."

Rakesh Sharma : "Commander, radiation levels are expected to increase rapidly. Anyone working outside must return to the protected habitat immediately."

Dr. Chiaki Mukai : "Some external power equipment is still operating. If we disconnect everything without preparation, we could lose valuable infrastructure."

Rakesh Sharma : "What is your recommendation"

M1 decision

Mission 2 — The Shelter Problem 6:00

Scene-2 : After the rescue operation, all five astronauts are safely inside the habitat.
the storm arrives

AstroBee: "Commander Commandor! The radiation monitoring system has detected a weakness in our habitat shielding. Our standard habitat walls cannot provide sufficient protection against the incoming solar particle event. We need to reinforce the crew shelter before the radiation reaches dangerous levels."

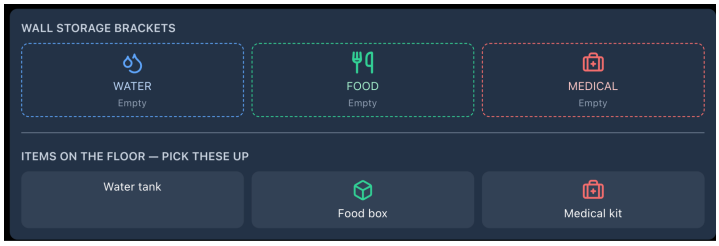
Rakesh Sharma : "Commander, we need additional shielding. The deepest protected section of our habitat offers better protection. We can also use regolith or water barriers to reduce radiation exposure."

Sunita Williams : "Remember that our stored water is essential for drinking and our greenhouse. If we use too much for shielding, we could create another problem."

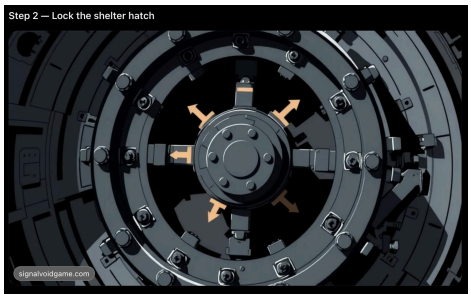
Rakesh Sharma : "What is your recommendation”

M2 decision

AstroBee: "Commandor, we have loose supplies near the shelter entrance. Pick up the water container, emergency food box, and medical kit. Place each one in its matching wall bracket. We need clear pathways before lockdown!"



AstroBee: "Now secure the shelter hatch rotate it 4 times and place on 75 degree angle"



Check Oxygen supply, Cabin circulation, CO_2 removal

Mission 3 — BlackOut 8:00

Scene-3 : normal operation control panel .
Camera moves outside to show the solar arrays.
One inverter status indicator changes from green to red.
The other inverter disconnects.
Power generation drops sharply.
An alarm appears: `SOLAR ARRAY POWER LOST`.

AstroBee: "Commander! Emergency alert! Our solar-array inverters have disconnected. The outpost is now running on emergency battery power. Its not safe to go outside and fix the solar array .We must manage our remaining electricity before the reserves run too low!"

Dr. Chiaki Mukai - Power & Energy Lead : "The main power supply is offline. Our emergency batteries can keep essential systems running, but we cannot operate everything. We need to make a decision quickly."

Sunita Williams - Hydroponics & Nutrition Lead : "The greenhouse depends on electricity for its grow lights and environmental controls. If we reduce its power for too long, our crops could suffer. But crew survival comes first."

Dr. Chiaki Mukai : "What is your recommendation”

M3 decision

Connect Emergency Battery → PDU.

Connect PDU → Oxygen Circulation.

Connect PDU → Habitat Heating.

Connect PDU → Emergency Communications.

Leave Decorative Lighting disconnected.

Mission 4 — Oxygen coutdown 11:00

Scene-4 : Show the radiation shelter's life-support control station.
An oxygen-reserve indicator starts falling.
An alert appears: `OXYGEN RESERVE LOW`.
AstroBee detects that oxygen production is insufficient.
The crew looks toward the life-support panel.

AstroBee: "Commander! Oxygen reserve is falling. Our current supply won't be enough unless we restore oxygen production or activate an alternative supply. We need to act quickly!"

O_2 Commando - ISRU & Oxygen Extraction Specialist : "We can use our emergency oxygen cylinders, produce fresh oxygen through water electrolysis, or isolate unused habitat modules to reduce unnecessary oxygen loss. But each option has a cost."

Dr. Chiaki Mukai- Power & Energy Lead : "Remember, Commander. Electrolysis needs electricity. Our emergency power is already limited, so watch the power reserve."

O2 commando : "What is your recommendation”

M4 decision

AstroBee: "Careful, Commandor! Electrolysis produces both oxygen and hydrogen. Never connect the hydrogen outlet to the breathable-air supply."

Connect Water Storage → Electrolysis Unit.

Connect Electrolysis Unit → Oxygen Separator.

Connect Oxygen Separator → Oxygen Storage Tank.

Connect Oxygen Storage Tank → Shelter Air Supply.

Connect the hydrogen outlet to its separate designated handling line. Never connect it to the oxygen line.

Mission 5 — Water Trap 13:00

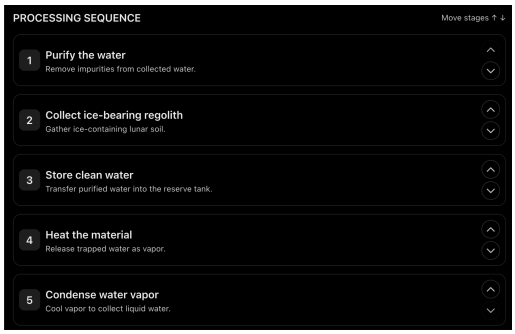
Scene-5 : The water purification system is operating at half capacity. The crew's clean-water reserve is falling, and the ice-processing system needs attention.

AstroBee: "Commander! Our clean-water production has dropped to 50%. We must secure enough water to keep the crew and greenhouse operating."

O_2 Commando - ISRU & Oxygen Extraction Specialist : "We have three possible strategies. Each one has a different impact on our resources."

O2 commando : "What is your recommendation"

M5 decision



sort in correct order.

Mission 6 — Last Harvest 16:00

Scene-6 :: The greenhouse nutrient circulation system has malfunctioned. Water and nutrients are no longer reaching the crops properly. The plants are under stress, and the crew's future food supply is at risk.

AstroBee: "Commander! We have a serious problem in the Plant Growth Lab. The nutrient circulation system has stopped working correctly. Our crops are beginning to deteriorate!"

Sunita Williams: "The plants need water, nutrients, light, and controlled temperature. If we don't stabilize the greenhouse soon, we could lose a major part of our future food supply."

Chiaki Mukai: "Power is still limited after the blackout. We must choose carefully how much energy to give the greenhouse."

Sunita Williams : "What is your recommendation”

M6 decision



AstroBee: "Greenhouse Light-Dark mode is broken. We need 16-8 hours consecutive light and dark cycle per day "

Mission 7 — Critical Failure 20:00

Scene-7 : A serious electrical fault has occurred at Power Bus Junction 4. The junction is overheating and producing electrical arcing. If the fault spreads, the entire outpost could lose power.

AstroBee: "Commander! Emergency alert! Power Bus Junction 4 is experiencing an electrical fault. The temperature is rising rapidly, and the power distribution network is becoming unstable!"

Chiaki Mukai: "We must isolate the damaged junction before the fault spreads. But be careful: disconnecting the wrong bus could shut down essential life-support systems."

O_2 Commando: "Our oxygen circulation and environmental systems depend on stable electrical power. We cannot afford to lose those circuits."

Chiaki Mukai: "What is your recommendation”

M7 decision

Mission 8 — The Last Signal 23:30

Average Earth–Moon distance: approximately 384,400 km.

Speed of light: approximately 299,792 km/s.

Travel time: $384,400 \div 299,792 \approx 1.28$ **seconds. Roughly 1.3s (two way 2.56 sec)**

Scene-8 : The 18-hour solar storm has passed. The crew is safe inside the shelter, but the outpost is still operating with limited power.

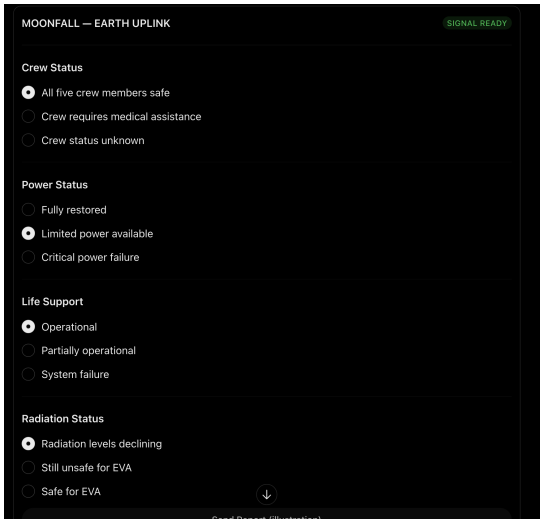
AstroBee: "Commander, the solar storm has passed. Radiation levels are decreasing, but our outpost systems are still recovering."

Rakesh Sharma: "The crew is safe, but we cannot assume the outpost is ready for normal operations. We need a complete system assessment."

Chiaki Mukai: "Our communication antenna is operational. We can finally send Mission Control our recovery report."

AstroBee:"Remember, Commander. Radio signals travel at the speed of light, but Earth cannot respond instantly. A message takes approximately 1.3 seconds to reach Earth, and another 1.3 seconds for a reply to return."

M8 decision



or chatbox